when the radius of the circular dimension changes from 10 to 1/10. And so, as far as total string energies go, there is no distinction between these different sizes for the circular dimension. just as the interchange of fitness-high/valves-low with valves-high/fitness-low is exactly compensated by an interchange of the number of shares held in each company, interchange of radius 10 and radius 1/10 is exactly compensated by the interchange of vibration and winding numbers. Moreover, while for simplicity we have focused on an initial radius of R = 10 and its reciprocal 1/10, the conclusions drawn are the same for any choice of the radius and its reciprocal.3

Tables 10.1 and 10.2 are incomplete for two reasons. First, as mentioned, we have listed only a few of the infinite possibilities for winding/vibration numbers that a string can assume. This, of course, poses no problem—we could make the tables as long as our patience allows and would find that the relation between them will continue to hold. Second, beyond winding energy, we have so far considered only energy contributions arising from the uniform-vibrational motion of a string. We should now include the ordinary vibrations as well, since these give additional contributions to the string's total energy and also determine the force charges it carries. The important point, however, is that investigations have revealed that these contributions do not depend on the size of the radius. Thus, even if we were to include these more detailed features of string attributes in Tables 10.1 and 10.2, the tables would still correspond exactly, since the ordinary vibrational contributions affect each table identically. We therefore conclude that the masses and the charges of particles in a Garden-hose universe with radius R are completely identical to those in a Garden-hose universe with radius 1/R. And since these masses and force charges govern fundamental physics, there is no way to distinguish physically these two geometrically distinct universes. Any experiment done in one such universe has a corresponding experiment that can be done in the other, leading to exactly the same results.

A Debate

George and Gracie, after being flattened out into two-dimensional beings, take up residence as physics professors in the Garden-hose universe. After setting up their competing laboratories, each claims to have determined the size of the circular dimension. Surprisingly,